Acme Cloud Platform Operations & Architecture Manual

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1. System Architecture & Infrastructure Overview

Acme Cloud operates distributed Kubernetes clusters running version 1.29 across three primary cloud regions: us-east (Northern Virginia), us-west (Oregon), and eu-central (Frankfurt). All production compute workloads are partitioned using namespace isolation and network policies.

The primary transactional database tier is deployed on PostgreSQL 16 managed via Patroni for automated high availability clustering. In the event of primary database node degradation or network partition, automated failover promotes a synchronous replica within a guaranteed window of 15 seconds.

Distributed session caching and global rate limiting are coordinated via a Redis 7.2 cluster topology. Asynchronous message streaming and audit event ingestion utilize an Apache Kafka 3.6 event bus deployed with a triple replication factor across independent availability zones.

2. Network Topology & Cryptographic Controls

All network ingress and egress traffic is enforced through TLS 1.3 protocol encryption. Legacy cipher suites such as TLS 1.0, 1.1, and 1.2 are rejected by ingress edge load balancers. Inter-service microservice communication is secured using mutual TLS (mTLS) with certificates rotated automatically every 30 days via cert-manager.

Data at rest across all storage volumes, persistent block stores, and object storage buckets is encrypted using AES-256 envelope encryption. Encryption keys are managed through customer-managed encryption keys (CMEK) backed by FIPS 140-2 Level 3 Hardware Security Modules.

3. Identity, Access Management & Authentication

Acme Cloud enforces strict Role-Based Access Control (RBAC) governed by four canonical system roles: Admin, Operator, Auditor, and Viewer. All user accounts must authenticate using hardware-backed FIDO2 WebAuthn security tokens or time-based one-time password (TOTP) multi-factor authentication.

Session tokens for active web and CLI sessions enforce a maximum lifespan of 60 minutes. Cryptographic refresh tokens are strictly limited to a duration of 24 hours and are revoked upon logout, password change, or IP address shift.

4. Data Retention, Compliance & Purge Schedules

Compliance guidelines dictate specific retention windows across distinct log categories. System audit trails and administrative event logs are retained for exactly 365 days within immutable, write-once-read-many (WORM) storage. Standard application debug logs and tracing metrics are permanently deleted after 90 days.

Right to Erasure requests submitted under GDPR Article 17 or CCPA regulations are scheduled and executed within a maximum turnaround of 30 calendar days. Upon execution, all associated user data records and cache entries are completely expunged across primary stores and secondary backup snapshots.

5. Service Level Agreements (SLAs) & Availability Commitments

Acme Cloud guarantees a 99.95% monthly uptime service level agreement for all active Enterprise Tier subscriptions. Uptime calculations exclude scheduled maintenance windows communicated at least 7 calendar days in advance.

Service credit compensations are structured on a graduated scale: if monthly uptime falls between 99.0% and 99.95%, customers are entitled to a 15% billing credit applied to their subsequent invoice. If monthly uptime falls below 99.0%, customers receive a 30% billing credit upon formal submission of an SLA reimbursement claim within 30 days.

6. Incident Severity Classification & Escalation Matrix

Incidents are categorized into three operational severity tiers: Severity 1 (Critical Incident) denotes complete outage or catastrophic data loss affecting multiple enterprise customers. Initial engineering response time is guaranteed under 15 minutes, with mandatory executive escalation to the Vice President of Engineering required within 30 minutes of incident declaration.

Severity 2 (Major Incident) designates degradation of core workflows where a viable operational workaround exists. Initial response SLA is capped at 60 minutes. Severity 3 (Minor Incident) represents cosmetic interface flaws or non-critical administrative latency, which carries a resolution SLA within 1 business day.

7. Subscription Pricing Tiers & Resource Quotas

Acme Cloud provides three standard subscription tiers tailored to organizational size: Starter Tier is billed at $49 per month and includes access for up to 5 users along with 50 GB of encrypted cloud storage. Professional Tier is billed at $199 per month, supporting up to 25 users and 500 GB of high-speed NVMe storage.

Enterprise Tier starts at $999 per month. Enterprise accounts include unlimited seat licenses, dedicated multi-region VPC peering, customized compliance reports, and 24/7 dedicated telephone support with designated technical account managers.

8. Disaster Recovery & Business Continuity

Disaster recovery protocols mandate strict objective metrics: Recovery Point Objective (RPO) is certified at 5 minutes for stateful transactional data through continuous write-ahead log (WAL) archiving. Recovery Time Objective (RTO) is certified at 30 minutes for comprehensive multi-region failover operations.

Simulated disaster recovery drills and unannounced failover exercises are executed semi-annually in May and November. Audit reports of drill results are made available to enterprise compliance officers within 14 days of completion.